Genome-wide fusion-hotspot cohort scan: msk_impact_50k_2026

544 of 3919 genes passed the >= 5-patient recurrence gate; 4 FDR-significant (q<0.05).

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Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
NTRK3	auto	ok	71	94.83	94.83	8.165e-09	yes	0.3943
ROS1	auto	ok	204	59.02	87.7	0.0002213	yes	0.3874
ETV6	auto	ok	213	71.11	75.56	0.002333	yes	0.4489
FLI1	auto	ok	121	61.86	5.932	0.008206	yes	0.5808
RET	curated	ok	219	75.26	92.27	0.09669	no	0.4345
FGFR2	auto	ok	188	79.41	84.56	0.09669	no	0.3373
TMPRSS2	auto	ok	1088	29.99	6.113	0.2147	no	0.5306
ERG	auto	ok	863	30.2	96.07	0.2147	no	0.4484
EWSR1	auto	ok	415	65.33	11.17	0.2147	no	0.2932
ALK	curated	ok	321	83.09	95.96	0.2147	no	0.4963
TP53	auto	ok	258	8.511	25.53	0.2147	no	0.2284
BRAF	curated	ok	247	84.36	91.06	0.2147	no	0.3299
EML4	auto	ok	224	87.89	53.36	0.2147	no	0.6667
FGFR3	auto	ok	194	48.68	93.42	0.2147	no	0.4924
TACC3	auto	ok	138	50	48.48	0.2147	no	0.5678
NOTCH1	auto	ok	115	23.4	59.57	0.2147	no	0.3398
NTRK1	curated	ok	113	41.03	82.05	0.2147	no	0.2923
NAB2	auto	ok	111	34.72	68.06	0.2147	no	0.3891
CDK12	auto	ok	99	16.28	37.21	0.2147	no	0.2757
TERT	auto	ok	92	0	68.42	0.2147	no	0.3993
MET	auto	ok	73	51.52	72.73	0.2147	no	0.285
POLE	auto	ok	73	18.18	31.82	0.2147	no	0.2045
STAT6	auto	ok	60	43.4	5.66	0.2147	no	0.3594
DNAJB1	auto	ok	54	64.1	2.564	0.2147	no	0.5432
RNF43	auto	ok	48	28.57	33.33	0.2147	no	0.2448
KIAA1549	auto	ok	44	90.91	50	0.2147	no	0.5302
POLD1	auto	ok	36	12.5	37.5	0.2147	no	0.2327
RBM10	auto	ok	36	42.86	71.43	0.2147	no	0.347
PRKACA	auto	ok	35	68.57	82.86	0.2147	no	0.6812
ATF1	auto	ok	30	70	90	0.2147	no	0.3942

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
BICC1	auto	ok	29	96.43	67.86	0.2147	no	0.4292
STAT5B	auto	ok	27	15.38	69.23	0.2147	no	0.2232
RPS6KA4	auto	ok	21	38.46	53.85	0.2147	no	0.2137
LMNA	auto	ok	17	46.15	38.46	0.2147	no	0.4387
AGK	auto	ok	14	92.86	14.29	0.2147	no	0.622
TPM3	auto	ok	14	78.57	64.29	0.2147	no	0.5362
NR4A3	auto	ok	7	57.14	71.43	0.2147	no	0.3723
EMID1	auto	ok	5	75	25	0.2147	no	0.4674
EGFR	auto	ok	466	50.91	74.55	0.2407	no	0.3219
CREB1	auto	ok	7	57.14	71.43	0.2407	no	0.3814
EZH1	auto	ok	30	0	18.18	0.2572	no	0.3722
TSC2	auto	ok	88	17.65	23.53	0.2769	no	0.2546
PRKD1	auto	ok	16	50	25	0.2769	no	0.3033
ASPSCR1	auto	ok	20	70	20	0.2847	no	0.4852
SEC16A	auto	ok	17	23.53	41.18	0.2847	no	0.3708
AGO2	auto	ok	49	26.09	30.43	0.3178	no	0.326
NCOR1	auto	ok	80	0	31.58	0.3288	no	0.3513
KDR	auto	ok	28	11.11	33.33	0.3288	no	0.2071
SLX4	auto	ok	28	26.32	57.89	0.3288	no	0.383
IKBKE	auto	ok	27	25	33.33	0.3288	no	0.315
CASP8	auto	ok	19	22.22	55.56	0.3288	no	0.2601
CDKN2B	auto	ok	40	16.67	16.67	0.3614	no	0.3909
CD74	auto	ok	74	58.21	88.06	0.3693	no	0.432
NOTCH2	auto	ok	92	3.571	50	0.3721	no	0.2303
CCDC6	auto	ok	53	92.45	1.887	0.3721	no	0.4276
DNMT3A	auto	ok	32	36.36	36.36	0.3721	no	0.3121
SUFU	auto	ok	17	25	25	0.3942	no	0.2274
RPTOR	auto	ok	43	23.08	15.38	0.4166	no	0.2673
BCAS3	auto	ok	18	6.25	43.75	0.4166	no	0.2759
PAX8	auto	ok	43	30	80	0.4549	no	0.2854
INPPL1	auto	ok	39	28.57	50	0.4687	no	0.2332
ACOXL	auto	ok	8	12.5	50	0.4902	no	0.3113
KIF5B	auto	ok	95	88.54	96.88	0.4986	no	0.6266
TOP1	auto	ok	31	0	50	0.5258	no	0.4077
WWOX	auto	ok	5	0	40	0.5258	no	0.4317
FGFR1	auto	ok	64	30.43	47.83	0.5434	no	0.2254
MAP3K1	auto	ok	42	10	30	0.5434	no	0.2556

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
CARM1	auto	ok	39	22.22	27.78	0.5434	no	0.2168
PRKAR1A	auto	ok	12	33.33	16.67	0.5434	no	0.2584
CREBBP	auto	ok	121	13.04	32.61	0.5449	no	0.2359
ERBB3	auto	ok	45	11.11	55.56	0.5772	no	0.219
STK11	auto	ok	97	4.348	30.43	0.5897	no	0.2657
CSDE1	auto	ok	31	0	60	0.5991	no	0.4667
TSC1	auto	ok	28	0	57.14	0.5991	no	0.4304
NRG1	auto	ok	16	53.33	86.67	0.6077	no	0.2663
FBXL20	auto	ok	8	28.57	28.57	0.6077	no	0.3592
FLT4	auto	ok	35	14.29	42.86	0.6241	no	0.2201
TRAF7	auto	ok	27	10	20	0.6484	no	0.2443
KMT2C	auto	ok	134	6.25	46.88	0.6719	no	0.2041
FH	auto	ok	29	6.25	12.5	0.6719	no	0.2761
ABL1	auto	ok	21	0	22.22	0.6719	no	0.3956
DGKH	auto	ok	7	0	28.57	0.6719	no	0.4272
ELF3	auto	ok	47	27.27	45.45	0.675	no	0.2607
RAD51B	auto	ok	16	50	50	0.675	no	0.3065
ARID1B	auto	ok	72	0	53.85	0.6823	no	0.4781
TFE3	auto	ok	45	56.76	78.38	0.6823	no	0.3513
AKT2	auto	ok	22	10	20	0.6823	no	0.2038
TTC28	auto	ok	15	16.67	58.33	0.6823	no	0.2223
GOPC	auto	ok	6	75	50	0.6823	no	0.4432
SND1	auto	ok	17	94.12	17.65	0.7043	no	0.3638
KREMEN1	auto	ok	7	28.57	42.86	0.7066	no	0.2453
FAT1	auto	ok	95	0	27.27	0.7068	no	0.4095
PLCG2	auto	ok	17	33.33	66.67	0.7173	no	0.2522
EXOC4	auto	ok	5	0	40	0.7205	no	0.5285
RAF1	auto	ok	23	25	50	0.7277	no	0.2143
ANKRD11	auto	ok	54	0	66.67	0.7606	no	0.4444
USP34	auto	ok	7	14.29	71.43	0.8029	no	0.4093
XPO1	auto	ok	31	8.333	41.67	0.8036	no	0.2217
MALT1	auto	ok	25	42.86	42.86	0.8092	no	0.2381
ESR1	auto	ok	30	23.08	38.46	0.8405	no	0.2334
PIK3CB	auto	ok	20	0	60	0.8405	no	0.4483
CD79A	auto	ok	13	0	37.5	0.8405	no	0.4167
SDHA	auto	ok	12	40	60	0.8405	no	0.3269
NOTCH3	auto	ok	101	3.846	50	0.8733	no	0.1881

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
FUBP1	auto	ok	29	18.18	54.55	0.8733	no	0.2052
ARID1A	auto	ok	148	12.9	35.48	0.8787	no	0.2097
CIC	auto	ok	68	0	42.86	0.8787	no	0.3363
KDM5A	auto	ok	44	30	50	0.8787	no	0.2332
ERBB2	auto	ok	88	15.91	43.18	0.897	no	0.2189
ATM	auto	ok	72	3.704	29.63	0.897	no	0.2108
BRCA1	auto	ok	66	7.407	48.15	0.897	no	0.1984
PPM1D	auto	ok	27	8.333	25	0.897	no	0.262
CACNA1A	auto	ok	7	14.29	42.86	0.897	no	0.2188
NELL1	auto	ok	5	20	80	0.897	no	0.3912
ARID5B	auto	ok	31	0	33.33	0.9186	no	0.4904
RB1	auto	ok	147	0	47.37	0.922	no	0.3984
KMT2A	auto	ok	65	19.05	47.62	0.922	no	0.1906
RAD21	auto	ok	38	44.44	55.56	0.922	no	0.3412
PICALM	auto	ok	5	0	60	0.922	no	0.4194
CSF3R	auto	ok	17	33.33	66.67	0.9248	no	0.2944
FOXA1	auto	ok	51	0	18.18	0.9569	no	0.5259
NF2	auto	ok	46	8.333	8.333	0.9569	no	0.2269
NOTCH4	auto	ok	44	0	68.75	0.9569	no	0.3691
SMARCA4	auto	ok	143	19.44	41.67	0.9926	no	0.2561
PAK1	auto	ok	25	10	10	0.9926	no	0.2267
CCND1	auto	ok	15	0	57.14	0.9926	no	0.4286
KMT2D	auto	ok	128	2.128	36.17	0.9997	no	0.1893
WT1	auto	ok	119	89.42	4.808	0.9997	no	0.6209
EP300	auto	ok	98	4.545	31.82	0.9997	no	0.2053
MGA	auto	ok	65	6.667	46.67	0.9997	no	0.2372
RICTOR	auto	ok	34	30	50	0.9997	no	0.2787
INSR	auto	ok	31	16.67	50	0.9997	no	0.2279
PDGFRB	auto	ok	16	40	40	0.9997	no	0.2707
MYD88	auto	ok	8	33.33	33.33	0.9997	no	0.2253
NF1	auto	ok	247	10.64	44.68	1	no	0.2742
APC	auto	ok	110	0	45.45	1	no	0.446
BRD4	auto	ok	86	3.125	34.38	1	no	0.1909
PTEN	auto	ok	86	22.22	22.22	1	no	0.2667
DOT1L	auto	ok	85	22.73	36.36	1	no	0.2478
PBRM1	auto	ok	85	5.556	50	1	no	0.2127
ZFHX3	auto	ok	85	0	14.29	1	no	0.5238

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
DNMT1	auto	ok	80	17.86	17.86	1	no	0.2299
BRCA2	auto	ok	79	8.333	58.33	1	no	0.214
ARID2	auto	ok	72	0	23.08	1	no	0.4156
CTNNB1	auto	ok	72	30	40	1	no	0.2134
KDM6A	auto	ok	71	25	75	1	no	0.08219
BAP1	auto	ok	67	16.67	33.33	1	no	0.2515
ATRX	auto	ok	64	28.57	42.86	1	no	0.281
SMAD4	auto	ok	62	0	16.67	1	no	0.4259
SPEN	auto	ok	61	0	26.32	1	no	0.357
CDH1	auto	ok	58	11.11	66.67	1	no	0.2045
PTPRD	auto	ok	57	12.5	37.5	1	no	0.2176
RTEL1	auto	ok	57	20.83	37.5	1	no	0.2173
PIK3R1	auto	ok	55	33.33	33.33	1	no	0.153
MDC1	auto	ok	52	7.143	57.14	1	no	0.2447
GLI1	auto	ok	51	19.05	57.14	1	no	0.2434
STAT3	auto	ok	51	8.333	8.333	1	no	0.1969
FANCA	auto	ok	49	15	35	1	no	0.239
MTOR	auto	ok	49	10	30	1	no	0.2128
ATR	auto	ok	48	0	28.57	1	no	0.3882
BRIP1	auto	ok	46	0	21.05	1	no	0.3595
SETD2	auto	ok	46	0	47.06	1	no	0.3671
NSD1	auto	ok	45	8.333	33.33	1	no	0.2471
PTPRS	auto	ok	44	7.143	35.71	1	no	0.1952
PIK3C2G	auto	ok	43	30	30	1	no	0.2381
B2M	auto	ok	42	0	0	1	no	0.6666
ERBB4	auto	ok	42	40	40	1	no	0.2314
DNMT3B	auto	ok	40	0	14.29	1	no	0.3882
KEAP1	auto	ok	40	0	25	1	no	0.3785
NFE2L2	auto	ok	40	0	20	1	no	0.5001
EZH2	auto	ok	39	14.29	7.143	1	no	0.2783
FOXP1	auto	ok	39	20	80	1	no	0.2861
NBN	auto	ok	38	0	33.33	1	no	0.4434
AXL	auto	ok	37	30	20	1	no	0.2136
PREX2	auto	ok	37	0	50	1	no	0.4466
BCOR	auto	ok	36	0	57.14	1	no	0.3882
DROSHA	auto	ok	36	42.86	14.29	1	no	0.3074
IGF1R	auto	ok	36	8.333	25	1	no	0.2019

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
TBX3	auto	ok	36	0	50	1	no	0.4012
KDM5C	auto	ok	33	30	60	1	no	0.2085
NCOA3	auto	ok	33	23.08	69.23	1	no	0.2059
PTPRT	auto	ok	33	33.33	50	1	no	0.221
RUNX1	auto	ok	33	16.67	33.33	1	no	0.2765
BCL2L11	auto	ok	32	21.43	21.43	1	no	0.3364
JAK3	auto	ok	32	0	33.33	1	no	0.392
PIK3R2	auto	ok	32	10	40	1	no	0.2632
RPS6KB2	auto	ok	32	33.33	0	1	no	0.2761
ERCC2	auto	ok	31	25	75	1	no	0.2651
PDGFRA	auto	ok	31	55.56	55.56	1	no	0.224
TP53BP1	auto	ok	31	18.18	54.55	1	no	0.2741
ETV1	auto	ok	30	52.38	0	1	no	0.3978
GNAS	auto	ok	30	25	0	1	no	0.2779
AXIN1	auto	ok	29	50	66.67	1	no	0.4256
CALR	auto	ok	29	44.44	0	1	no	0.2806
CBL	auto	ok	29	25	50	1	no	0.3399
JAK1	auto	ok	29	12.5	0	1	no	0.2064
NTRK2	auto	ok	29	0	40	1	no	0.4405
AXIN2	auto	ok	28	0	75	1	no	0.4466
RARA	auto	ok	28	28.57	35.71	1	no	0.2158
TEK	auto	ok	28	0	55.56	1	no	0.3709
TET1	auto	ok	28	14.29	28.57	1	no	0.2135
MAP2K4	auto	ok	27	14.29	0	1	no	0.3107
SMAD3	auto	ok	27	11.11	55.56	1	no	0.2404
IRS2	auto	ok	26	11.11	77.78	1	no	0.2554
SOX9	auto	ok	26	0	20	1	no	0.4441
UPF1	auto	ok	26	33.33	50	1	no	0.235
CARD11	auto	ok	25	0	28.57	1	no	0.4304
FLT1	auto	ok	25	33.33	33.33	1	no	0.1589
ASXL2	auto	ok	24	28.57	57.14	1	no	0.2761
DIS3	auto	ok	24	16.67	16.67	1	no	0.3189
TP63	auto	ok	24	28.57	14.29	1	no	0.2747
AKT1	auto	ok	23	0	42.86	1	no	0.5238
MAP3K13	auto	ok	23	0	20	1	no	0.4385
MDM2	auto	ok	23	23.53	47.06	1	no	0.2584
MED12	auto	ok	23	0	33.33	1	no	0.4169

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
SMARCB1	auto	ok	23	0	60	1	no	0.5993
STAT5A	auto	ok	23	7.692	0	1	no	0.204
TCF7L2	auto	ok	23	0	50	1	no	0.5
EIF4A2	auto	ok	22	0	33.33	1	no	0.4012
PARP1	auto	ok	22	10	40	1	no	0.2584
PIK3CD	auto	ok	22	12.5	50	1	no	0.2195
PLK2	auto	ok	22	28.57	57.14	1	no	0.3264
PPP2R1A	auto	ok	22	50	33.33	1	no	0.2948
PRKCI	auto	ok	22	33.33	50	1	no	0.2899
RAD54L	auto	ok	22	0	60	1	no	0.4194
FLT3	auto	ok	21	0	20	1	no	0.4441
MAP2K2	auto	ok	21	28.57	14.29	1	no	0.2448
SMARCD1	auto	ok	21	12.5	50	1	no	0.2672
BLM	auto	ok	20	0	50	1	no	0.4135
FLCN	auto	ok	20	50	40	1	no	0.2682
MSH6	auto	ok	20	0	66.67	1	no	0.4012
PIK3R3	auto	ok	20	33.33	0	1	no	0.1419
SMYD3	auto	ok	20	25	25	1	no	0.1125
SRC	auto	ok	20	25	0	1	no	0.2485
CHEK2	auto	ok	19	16.67	16.67	1	no	0.2764
CSF1R	auto	ok	19	0	33.33	1	no	0.3709
GRIN2A	auto	ok	19	0	16.67	1	no	0.4012
LATS1	auto	ok	19	16.67	33.33	1	no	0.2232
PTCH1	auto	ok	19	0	20	1	no	0.4317
YAP1	auto	ok	19	20	20	1	no	0.2861
FGFR4	auto	ok	18	11.11	44.44	1	no	0.2053
PGR	auto	ok	18	14.29	57.14	1	no	0.2665
VTCN1	auto	ok	18	20	60	1	no	0.288
EGFL7	auto	ok	17	50	25	1	no	0.4086
ERCC5	auto	ok	17	0	37.5	1	no	0.5
LDLR	auto	ok	17	20	53.33	1	no	0.3759
SDC4	auto	ok	17	82.35	0	1	no	0.4731
TAP1	auto	ok	17	0	42.86	1	no	0.4286
TGFBR1	auto	ok	17	25	75	1	no	0.2714
CTCF	auto	ok	16	0	50	1	no	0.5
CUL1	auto	ok	16	35.71	0	1	no	0.3837
RECQL	auto	ok	16	0	0	1	no	0.4713

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
SMO	auto	ok	16	0	12.5	1	no	0.4032
SUZ12	auto	ok	16	0	20	1	no	0.4194
CCNE1	auto	ok	15	20	0	1	no	0.2861
MSH3	auto	ok	15	0	50	1	no	0.5
TNFRSF14	auto	ok	15	25	0	1	no	0.2625
CDK4	auto	ok	14	33.33	0	1	no	0.1183
E2F3	auto	ok	14	0	20	1	no	0.4194
FOXO1	auto	ok	14	50	0	1	no	0.3721
MPL	auto	ok	14	0	83.33	1	no	0.4665
NUF2	auto	ok	14	40	10	1	no	0.2888
NUP93	auto	ok	14	0	20	1	no	0.4194
SESN3	auto	ok	14	20	0	1	no	0.1214
SF3B1	auto	ok	14	20	40	1	no	0.2872
EZR	auto	ok	13	61.54	0	1	no	0.4151
MST1R	auto	ok	13	20	40	1	no	0.253
NPM1	auto	ok	13	50	50	1	no	0.3172
TRAPPC9	auto	ok	13	23.08	0	1	no	0.4889
YES1	auto	ok	13	16.67	16.67	1	no	0.2443
AKT3	auto	ok	12	33.33	33.33	1	no	0.267
CEBPA	auto	ok	12	0	0	1	no	0.4444
DUSP4	auto	ok	12	0	33.33	1	no	0.4135
FYN	auto	ok	12	66.67	66.67	1	no	0.1422
IDH1	auto	ok	12	25	0	1	no	0.2625
PAX5	auto	ok	12	33.33	66.67	1	no	0.186
PIK3C3	auto	ok	12	100	100	1	no	0.3772
PIK3CA	auto	ok	12	0	80	1	no	0.5222
ERCC4	auto	ok	11	20	40	1	no	0.2872
FGF19	auto	ok	11	0	0	1	no	0.4888
GSK3B	auto	ok	11	50	0	1	no	0.2625
MKRN1	auto	ok	11	63.64	0	1	no	0.3815
MLH1	auto	ok	11	50	50	1	no	0.2625
PHF7	auto	ok	11	50	0	1	no	0.2625
RAD51D	auto	ok	11	50	0	1	no	0.2625
RELA	auto	ok	11	33.33	100	1	no	0.426
SOS1	auto	ok	11	33.33	0	1	no	0.1465
TACC2	auto	ok	11	62.5	100	1	no	0.5485
BMPR1A	auto	ok	10	100	0	1	no	0.3772

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
CDK8	auto	ok	10	25	0	1	no	0.1472
CUL3	auto	ok	10	50	0	1	no	0.2599
EPAS1	auto	ok	10	0	20	1	no	0.4194
IKZF3	auto	ok	10	25	62.5	1	no	0.2719
MDM4	auto	ok	10	25	50	1	no	0.3137
RXRA	auto	ok	10	33.33	66.67	1	no	0.1419
TRIM24	auto	ok	10	88.89	0	1	no	0.3976
ARAF	auto	ok	9	50	50	1	no	0.2082
LRP1	auto	ok	9	0	20	1	no	0.4347
NTHL1	auto	ok	9	33.33	66.67	1	no	0.1961
PPARG	auto	ok	9	50	50	1	no	0.2072
RAD51	auto	ok	9	66.67	66.67	1	no	0.2773
SESN1	auto	ok	9	0	0	1	no	0.4194
STK40	auto	ok	9	33.33	0	1	no	0.1953
ETV4	auto	ok	8	33.33	16.67	1	no	0.3893
HNF1A	auto	ok	8	0	20	1	no	0.4441
IKZF1	auto	ok	8	100	0	1	no	0.3772
IRF4	auto	ok	8	0	20	1	no	0.4667
KRAS	auto	ok	8	33.33	0	1	no	0.1856
MAPK1	auto	ok	8	50	0	1	no	0.2599
MCL1	auto	ok	8	0	50	1	no	0.4538
MSI1	auto	ok	8	20	40	1	no	0.2192
MTAP	auto	ok	8	25	25	1	no	0.4351
PDE4D	auto	ok	8	16.67	66.67	1	no	0.3623
PTK2	auto	ok	8	25	50	1	no	0.276
SLC25A21	auto	ok	8	0	12.5	1	no	0.4466
STRN	auto	ok	8	25	0	1	no	0.5381
CDK6	auto	ok	7	50	0	1	no	0.2599
CRKL	auto	ok	7	0	50	1	no	0.6442
DCUN1D1	auto	ok	7	66.67	0	1	no	0.1606
DNM2	auto	ok	7	28.57	42.86	1	no	0.2286
GAB2	auto	ok	7	20	20	1	no	0.2861
IGF2	auto	ok	7	50	75	1	no	0.2614
ST7	auto	ok	7	14.29	0	1	no	0.437
TACC1	auto	ok	7	25	75	1	no	0.2625
CAMTA1	auto	ok	6	20	0	1	no	0.1241
CDKN2C	auto	ok	6	33.33	66.67	1	no	0.1594

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
CPM	auto	ok	6	40	0	1	no	0.1385
CTNNA3	auto	ok	6	60	0	1	no	0.2525
DTNB	auto	ok	6	16.67	0	1	no	0.3383
EIF3H	auto	ok	6	60	0	1	no	0.529
EPHX3	auto	ok	6	16.67	0	1	no	0.2625
NOS1AP	auto	ok	6	0	16.67	1	no	0.4434
SMARCA2	auto	ok	6	33.33	66.67	1	no	0.1424
SOX2	auto	ok	6	0	75	1	no	0.5009
TANC2	auto	ok	6	16.67	50	1	no	0.2263
TFG	auto	ok	6	100	100	1	no	0.3246
ZNRF3	auto	ok	6	33.33	66.67	1	no	0.4336
ATE1	auto	ok	5	33.33	0	1	no	0.3752
BCL2L1	auto	ok	5	50	50	1	no	0.2093
CNTNAP2	auto	ok	5	66.67	0	1	no	0.267
CREM	auto	ok	5	20	100	1	no	0.2602
CTTNBP2	auto	ok	5	0	20	1	no	0.5285
EIF4E	auto	ok	5	50	0	1	no	0.2599
FBXL7	auto	ok	5	50	75	1	no	0.4249
GPATCH8	auto	ok	5	25	50	1	no	0.3093
HCN1	auto	ok	5	100	0	1	no	0.3772
LRP1B	auto	ok	5	50	50	1	no	0.2008
MAD1L1	auto	ok	5	40	0	1	no	0.2382
PRKAG2	auto	ok	5	20	40	1	no	0.2774
RAC2	auto	ok	5	50	0	1	no	0.2599
SBNO2	auto	ok	5	0	75	1	no	0.4466
SDHAF2	auto	ok	5	100	0	1	no	0.4933
SFPQ	auto	ok	5	25	75	1	no	0.4008
SHANK2	auto	ok	5	0	25	1	no	0.4466
SMARCE1	auto	ok	5	100	0	1	no	0.4933
SMURF2	auto	ok	5	25	75	1	no	0.2569
SPTBN1	auto	ok	5	50	0	1	no	0.3932
THSD4	auto	ok	5	25	25	1	no	0.2439
TPR	auto	ok	5	40	0	1	no	0.5142
CDKN2A	auto	ok	185	4.762	0			0.2857
KMT2B	unresolved	failed	68					
NSD3	auto	ok	58	14.29	0			0.1429
RECQL4	unresolved	failed	54					

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
PRKN	auto	ok	40	50	0			0.25
NSD2	auto	ok	37	33.33	0			0.3333
MAP3K14	unresolved	failed	36					
ASXL1	auto	ok	35	0	77.78			0.2222
TCF3	unresolved	failed	35					
RASA1	auto	ok	32	0	50			0.5
STAG2	auto	ok	31	0	0			0.5
AR	auto	ok	30	0	50			0.5
CDC73	auto	ok	28	0	50			0.5
JAK2	auto	ok	27	0	33.33			0.3016
CDKN1B	auto	ok	24	0	0			0.3016
MSH2	auto	ok	23	0	0			0.2731
NFKBIA	auto	ok	23	0	100			1
MEN1	auto	ok	22	0	0			0.3016
MYC	auto	ok	22	0	0			0.3333
TET2	auto	ok	21	0	66.67			0.5516
ERF	unresolved	failed	20					
KIT	auto	ok	20	0	0			0.4266
CDKN2B-AS1	unresolved	failed	19					
EPHA3	auto	ok	19	0	0			0.4266
NEGR1	auto	ok	19	0	66.67			0.3333
PALB2	auto	ok	19	0	0			0.2
SEPTIN14	unresolved	failed	19					
BBC3	auto	ok	18	50	0			0.375
GATA3	auto	ok	18	0	0			0.5
AMER1	auto	ok	17	0	33.33			0.3333
COP1	auto	ok	17	50	0			0.5
DDR2	auto	ok	17	0	0			0.2016
EPHA5	auto	ok	17	0	33.33			0.5516
INPP4B	auto	ok	17	100	0			1
RAD50	auto	ok	17	0	33.33			0.3016
BCL6	auto	ok	16	0	66.67			0.3016
DICER1	auto	ok	16	0	0			0.3856
ERRFI1	auto	ok	16	0	0			0.6667
LATS2	auto	ok	16	0	0			0.8016
MITF	auto	ok	16	0	100			1
MSI2	auto	ok	16	0	50			0.4266

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
NCOA4	auto	failed	16					
RAD52	auto	ok	16	0	100			0.8016
TGFBR2	auto	ok	16	0	50			0.5
TRAP1	unresolved	failed	16					
CREB3L1	unresolved	failed	15					
INPP4A	auto	ok	15	14.29	0			0.2857
SRP19	auto	ok	15	0	0			1
GPS2	auto	ok	14	0	0			0.5
HGF	auto	ok	14	0	50			0.4606
LINC00114	unresolved	failed	14					
RIT1	auto	ok	14	0	0			0.3596
TAP2	auto	ok	14	0	0			0.5
TRAF2	unresolved	failed	14					
CYLD	auto	ok	13	0	0			1
DAXX	auto	ok	13	0	0			1
MAP2K1	auto	ok	13	0	50			0.25
MEF2B	unresolved	failed	13					
MRE11	auto	ok	13	0	20			0.2
PAK5	auto	ok	13	50	0			0.5
RAD51C	auto	ok	13	0	0			0.1766
SPOP	auto	ok	13	0	50			0.4266
BIRC3	auto	ok	12	0	66.67			0.3016
BTK	auto	ok	12	0	0			1
CBFB	auto	ok	12	0	0			1
EED	auto	ok	12	0	0			0.4266
IRS1	auto	ok	12	0	0			0.5
RAB35	auto	ok	12	0	0			0.3356
SMAD2	auto	ok	12	0	0			0.8016
VEGFA	auto	ok	12	0	0			0.4606
ALOX12B	auto	ok	11	0	0			0.4266
BABAM1	auto	ok	11	0	0			0.5
BCL2	auto	ok	11	0	0			1
EPHA7	auto	ok	11	0	50			0.4266
FANCC	auto	ok	11	0	0			0.6096
FBXW7	auto	ok	11	0	0			0.3016
IDH2	auto	ok	11	0	0			0.4266
SRSF2	auto	ok	11	0	100			1

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
TNFAIP3	auto	ok	11	0	0			0.3016
AURKA	auto	ok	10	0	33.33			0.5516
CCND2	auto	ok	10	0	60			0.3516
EPHB1	auto	ok	10	0	50			0.5
H3C2	unresolved	failed	10					
MAPK3	auto	ok	10	0	0			0.8016
PIK3CG	auto	ok	10	0	0			
PMS1	auto	ok	10	0	0			
PTPN11	auto	ok	10	0	0			0.3333
SH2B3	auto	ok	10	0	25			0.25
SOX17	auto	ok	10	0	0			0.5856
SYK	auto	ok	10	0	0			1
CHEK1	auto	ok	9	0	0			0.4266
GATA1	auto	ok	9	0	0			1
IFNGR1	auto	ok	9	0	33.33			0.3016
MAPKAP1	auto	ok	9	0	100			0.8016
PPP6C	auto	ok	9	0	0			1
PRDM14	auto	ok	9	0	0			
REL	auto	ok	9	0	0			
SHQ1	auto	ok	9	0	50			0.5
AKAP8	auto	ok	8	0	0			1
AURKB	auto	ok	8	0	33.33			0.3333
BARD1	auto	ok	8	0	0			
BCL2L14	auto	ok	8	50	0			0.75
CCND3	auto	ok	8	0	0			0.8356
CD79B	auto	ok	8	0	33.33			0.3016
ERCC3	auto	ok	8	0	33.33			0.3016
FGF3	auto	ok	8	0	0			0.3333
GNA11	auto	ok	8	0	0			1
ICOSLG	auto	ok	8	0	0			
LYN	auto	ok	8	0	0			0.2391
NBPF20	unresolved	failed	8					
PDCD1	auto	ok	8	0	0			1
RRAS	auto	ok	8	0	33.33			0.3333
XRCC2	auto	ok	8	0	0			1
ACP3	auto	ok	7	71.43	0			0.4286
AP1B1	auto	ok	7	0	0			1

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
ASIC2	auto	ok	7	16.67	0			0.3333
ATP1B2	auto	ok	7	0	0			1
DCBLD1	auto	ok	7	0	0			
EPCAM	auto	ok	7	0	0			0.5
H3C4	auto	ok	7	0	0			
INHA	auto	ok	7	0	0			1
JUN	auto	ok	7	0	0			
KLF4	auto	ok	7	0	0			
NKX3-1	auto	ok	7	0	0			1
PDPK1	auto	ok	7	0	100			1
RAB11FIP4	unresolved	failed	7					
RYBP	auto	ok	7	0	0			
SDHC	auto	ok	7	0	66.67			0.3016
SHOC2	auto	ok	7	0	100			1
TENT5C	auto	ok	7	0	0			
AGAP3	unresolved	failed	6					
CD274	auto	ok	6	0	0			1
CD276	auto	ok	6	0	0			
CDH3	unresolved	failed	6					
FGF4	auto	ok	6	0	50			0.5
GALNT11	auto	ok	6	0	33.33			1
GRB7	auto	ok	6	0	20			0.8
H1-2	auto	ok	6	0	0			1
H3-3B	auto	ok	6	0	0			0.5
H3C6	unresolved	failed	6					
HOXB13	auto	ok	6	0	0			0.5

Showing the first 500 of 544 scanned genes.

Honorable mentions: highly ranked non-FDR-significant genes worth human review

Did NOT survive genome-wide FDR correction -- ranked by raw Fisher p-value among non-significant genes. Not a claim of significance.

Rank	Gene	Fisher p	Best q	N events	In-frame%	Domain-ret.%
1	RET	0.0004197	0.09669	194	75.26	92.27
2	FGFR2	0.0004247	0.09669	136	79.41	84.56

Rank	Gene	Fisher p	Best q	N events	In-frame%	Domain-ret.%
3	ALK	0.001917	0.2147	272	83.09	95.96
4	EGFR	0.01145	0.2407	55	50.91	74.55
5	BRAF	0.01337	0.2147	179	84.36	91.06
6	FGFR3	0.01948	0.2147	152	48.68	93.42
7	CDKN2B	0.02222	0.3614	12	16.67	16.67
8	IKBKE	0.02424	0.3288	12	25	33.33
9	TMPRSS2	0.0292	0.2147	867	29.99	6.113
10	NTRK1	0.04826	0.2147	78	41.03	82.05
11	INPPL1	0.04895	0.4687	14	28.57	50
12	CDK12	0.0677	0.2147	43	16.28	37.21
13	NAB2	0.1126	0.2147	72	34.72	68.06
14	KDM5A	0.119	0.8787	10	30	50
15	FH	0.1333	0.6719	16	6.25	12.5

Figures

manhattan